

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

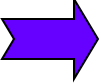
Chap. 4 Open-Loop Discrete-Time Systems

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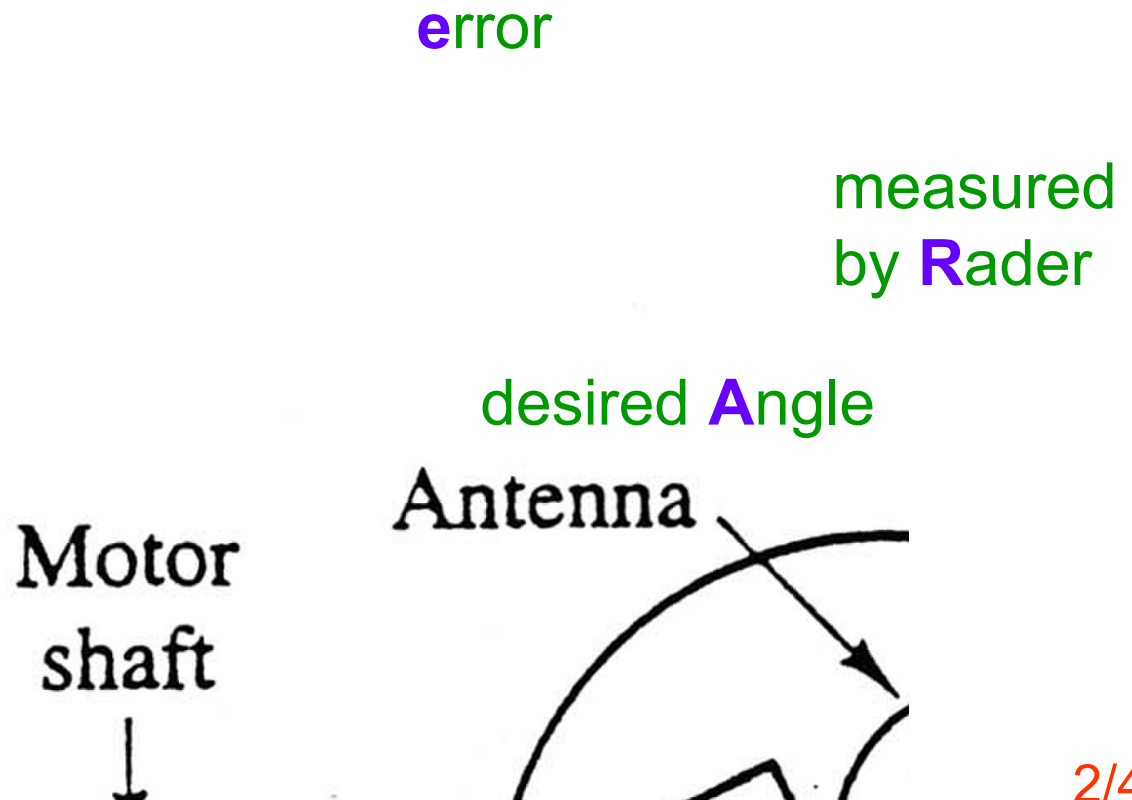
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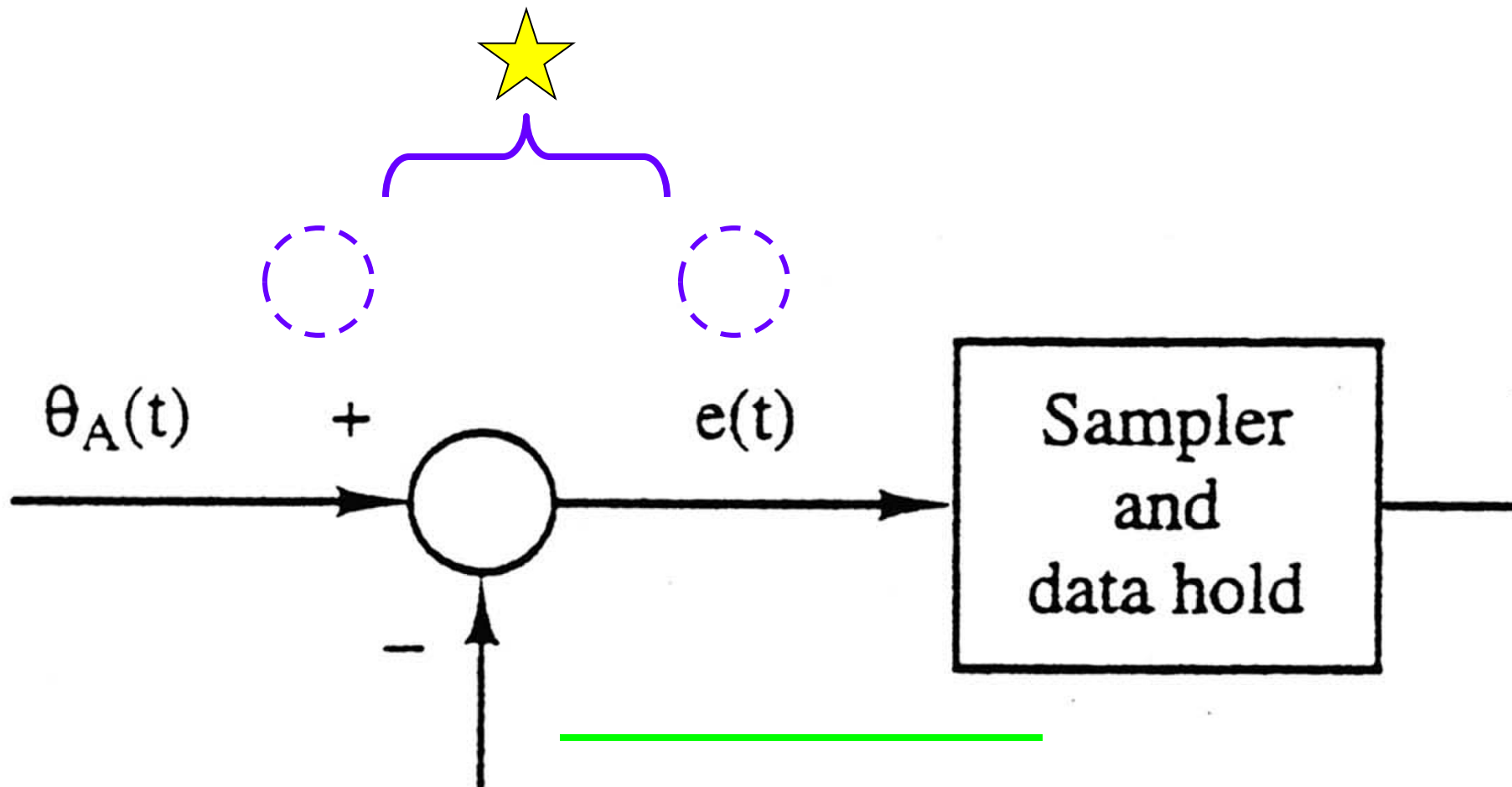
Chap. 7 Stability Analysis Techniques

Chap. 8 Digital Controller Design

 Determine the effects of sampling C-T signals.

3.2 Sampled-Data Control Systems





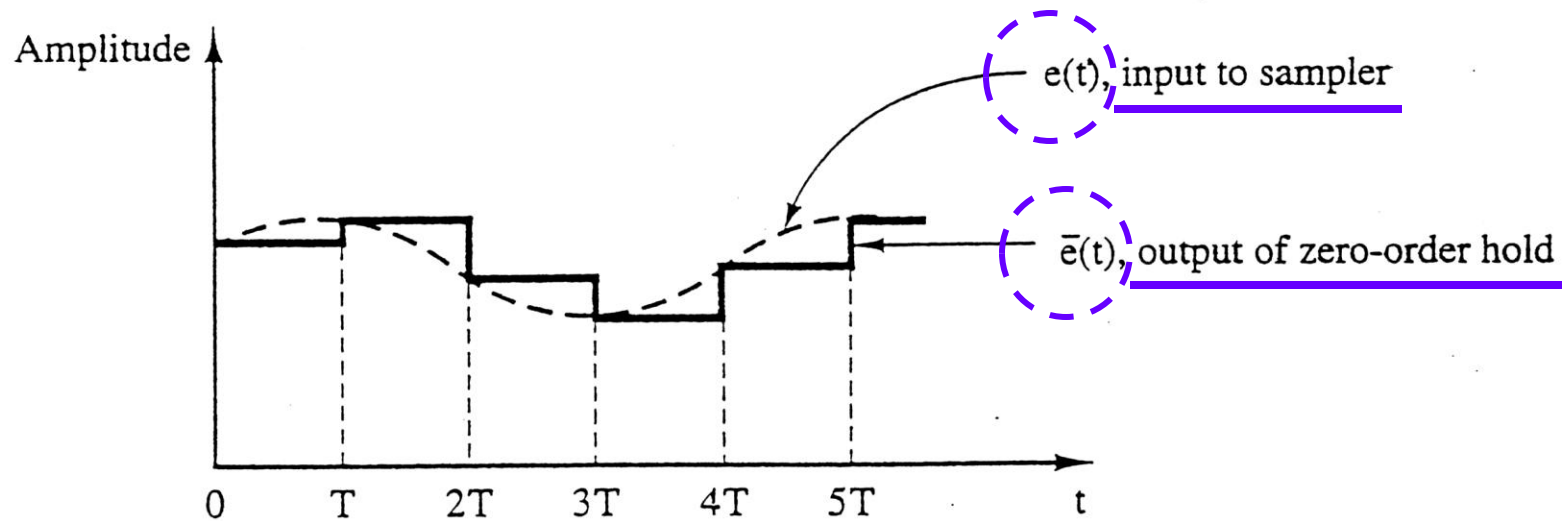
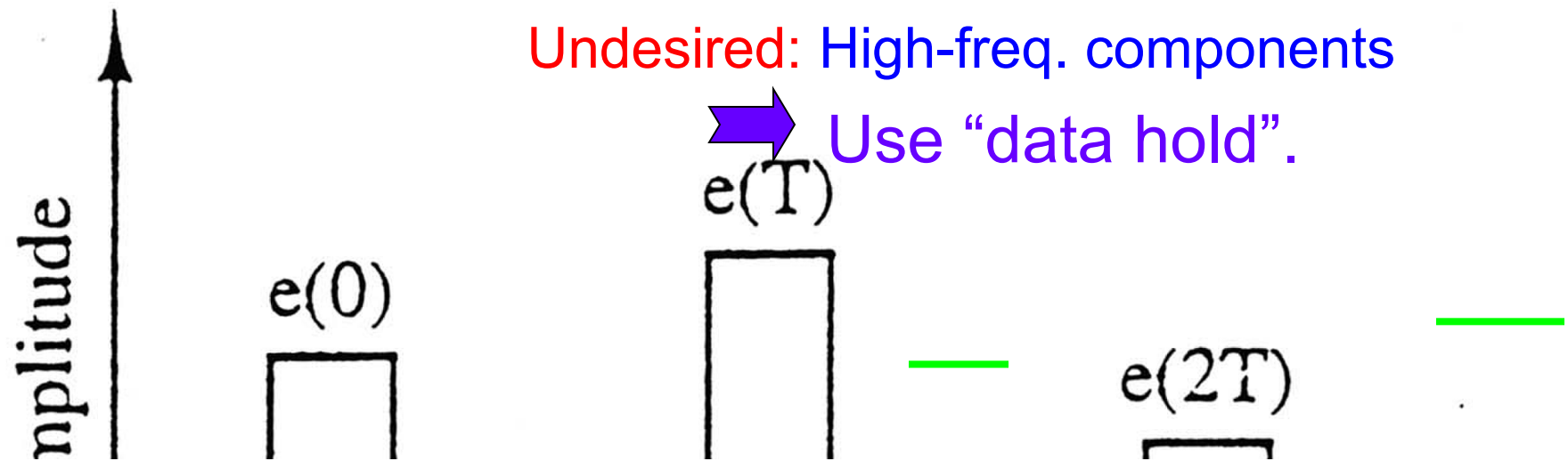
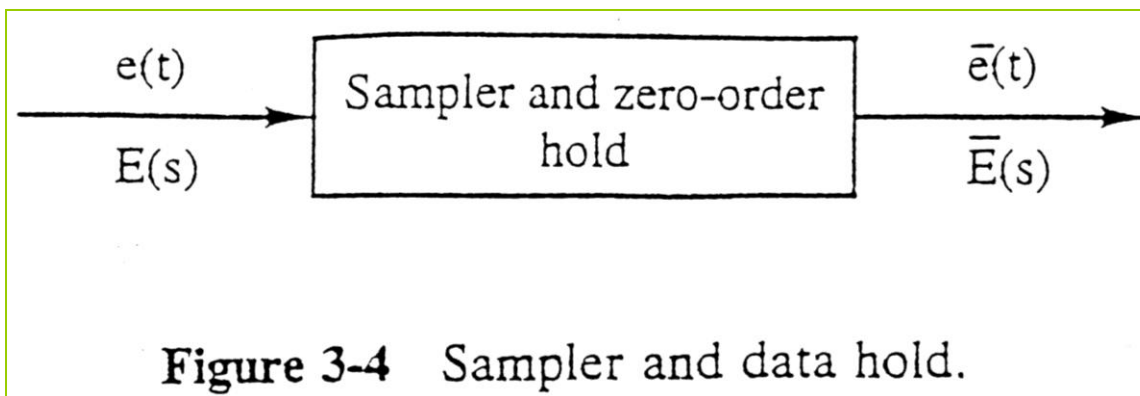


Figure 3-3 Input and output signals of sampler/data hold.



The signal $\bar{e}(t)$ can be expressed as

$$\begin{aligned} \bar{e}(t) = & e(0)[u(t) - u(t - T)] + e(T)[u(t - T) - u(t - 2T)] \\ & + e(2T)[u(t - 2T) - u(t - 3T)] + \dots \end{aligned} \quad (3-1)$$

where $u(t)$ is the unit-step function. The Laplace transform of $\bar{e}(t)$ is $\bar{E}(s)$, given by

$$\begin{aligned} \bar{E}(s) = & e(0) \left[\frac{1}{s} - \frac{\epsilon^{-Ts}}{s} \right] + e(T) \left[\frac{\epsilon^{-Ts}}{s} - \frac{\epsilon^{-2Ts}}{s} \right] \\ & + e(2T) \left[\frac{\epsilon^{-2Ts}}{s} - \frac{\epsilon^{-3Ts}}{s} \right] + \dots \end{aligned} \quad \boxed{\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}}$$

$$\begin{aligned} & = \left[\frac{1 - \epsilon^{-Ts}}{s} \right] [e(0) + e(T)\epsilon^{-Ts} + e(2T)\epsilon^{-2Ts} + \dots] \\ & = \left[\sum_{n=0}^{\infty} e(nT)\epsilon^{-nTs} \right] \left[\frac{1 - \epsilon^{-Ts}}{s} \right] \end{aligned} \quad (3-2)$$

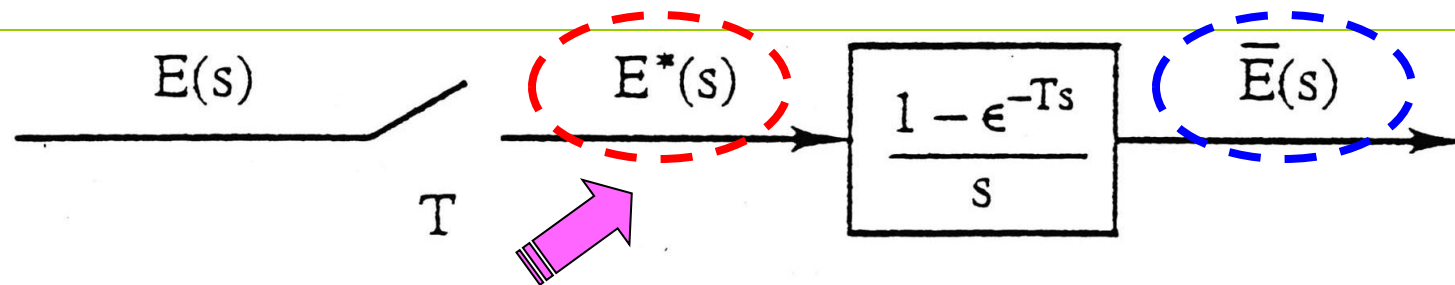


Figure 3-5 Representation of sampler and data hold.

The first factor in the last expression in (3-2) is seen to be a function of the input signal $e(t)$ and the sampling period T . The second factor is seen to be independent of $e(t)$ and therefore can be considered to be a transfer function. Thus the sampler/

hold operation can be represented as shown in Figure 3-4. The function $E^*(s)$, called the starred transform, is defined as

$$E^*(s) = \sum_{n=0}^{\infty} e(nT) e^{-nTs} \quad (3-3)$$

- $E^*(s)$ is NOT a physical system.
- No TF exists for the sampler.

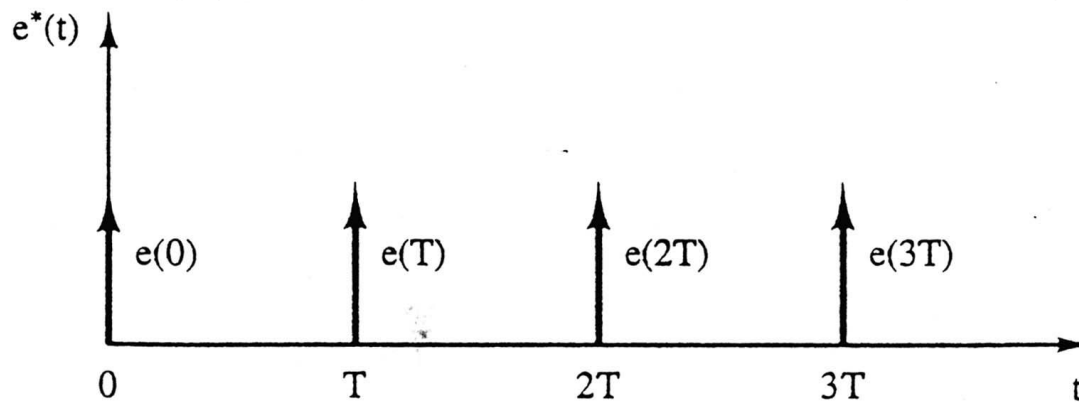
3.3 Ideal Sampler

The inverse Laplace transform of $E^*(s)$ is, from (3-3),

$$\begin{aligned} e^*(t) = \mathcal{L}^{-1}[E^*(s)] &= e(0)\delta(t) + e(T)\delta(t - T) \\ &+ e(2T)\delta(t - 2T) + \dots \end{aligned} \quad (3-4)$$

where $\delta(t)$ is the unit impulse function occurring at $t = 0$. Then $e^*(t)$ is a train of

impulse functions whose weights are equal to the values of the signal at the instants of sampling. Thus $e^*(t)$ can be represented as shown in Figure 3-6, since the impulse function has infinite amplitude at the instant it occurs. Note again that $e^*(t)$ is not a physical signal.



$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}$$

$$\mathcal{L}\{\delta(t)\} = 1$$

Figure 3-6 Representation of $e^*(t)$.

$$\delta_T(t) = \sum_{n=0}^{\infty} \delta(t - nT)$$



signal

$e(t)$

$e^*(t)$

A problem arises in the definition of the ideal sampler output in (3-4) if $e(t)$ has a discontinuity at $t = kT$. For example, if $e(t)$ is a unit-step function, what value is used for $e(0)$ in (3-4)? In order to be consistent in the consideration of discontinuous signals, the output signal of an ideal sampler is *defined* as follows:

Definition. The output signal of an ideal sampler is defined as the signal whose Laplace transform is

$$E^*(s) = \sum_{n=0}^{\infty} e(nT) \epsilon^{-nTs} \quad (3-7)$$

where $e(t)$ is the input signal to the sampler. If $e(t)$ is discontinuous at $t = kT$, where k is an integer, then $e(kT)$ is taken to be $e(kT^+)$. The notation $e(kT^+)$ indicates the value of $e(t)$ as t approaches kT from the right (i.e., at $t = kT + \epsilon$, where ϵ is made arbitrarily small).

3.4 Evaluation of $E^*(s)$

$E^*(s)$, as defined by (3-7), has limited usefulness in analysis because it is expressed as an infinite series. However, for many useful time functions, $E^*(s)$ can be expressed in closed form. In addition, there is a third form of $E^*(s)$ that is useful. These two additional forms of $E^*(s)$ will now be investigated.

As was seen in Section 3.3, we can express the inverse Laplace transform of $E^*(s)$ as [see (3-6)]

$$e^*(t) = e(t)\delta_T(t) \quad (3-9)$$

If we then take the Laplace transform of $e^*(t)$ using the complex convolution integral [3], we can derive two additional expressions for $E^*(s)$. These derivations are given in Appendix III. The resultant expressions are

Residues: (2-33) on P46

Closed Form $E^*(s) = \sum_{\substack{\text{at poles} \\ \text{of } E(\lambda)}} \left[\text{residues of } E(\lambda) \frac{1}{1 - e^{-T(s - \lambda)}} \right] \quad (3-10)$
(Sec. 3.4)

Properties $E^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \underline{E(s + jn\omega_s)} + \frac{e(0)}{2} \text{ periodic} \quad (3-11)$
(Sec. 3.6)

where ω_s is the radian sampling frequency, that is, $\omega_s = 2\pi/T$. The residue of a function is defined in Section 2.6. The expression (3-10) is useful in generating tables for the starred transform $E^*(s)$, and expression (3-11) will prove to be useful in analysis, as illustrated in the next section. Some examples that illustrate the use of (3-10) will now be given.

Example 3.3

Example 3.3

Determine $E^*(s)$ given that

$$E(s) = \frac{1}{(s+1)(s+2)}$$

From (3-10),

$$E(\lambda) \frac{1}{1 - e^{-T(s-\lambda)}} = \frac{1}{(\lambda+1)(\lambda+2)(1 - e^{-T(s-\lambda)})}$$

Then

$$\begin{aligned} E^*(s) &= \sum_{\text{poles of } E(\lambda)} \left[\text{residues of } E(\lambda) \frac{1}{1 - e^{-T(s-\lambda)}} \right] \\ &= \frac{1}{(\lambda+2)(1 - e^{-T(s-\lambda)})} \Big|_{\lambda=-1} + \frac{1}{(\lambda+1)(1 - e^{-T(s-\lambda)})} \Big|_{\lambda=-2} \\ &= \frac{1}{1 - e^{-T(s+1)}} - \frac{1}{1 - e^{-T(s+2)}} \end{aligned}$$

Example 3.4

Example 3.4

We wish to determine the starred transform of $e(t) = \sin \beta t$. The corresponding $E(s)$ is

$$E(s) = \frac{\beta}{s^2 + \beta^2} = \frac{\beta}{(s - j\beta)(s + j\beta)}$$

$E^*(s)$ can be evaluated from the expression

$$\begin{aligned} E^*(s) &= \sum_{\text{poles of } E(\lambda)} \left[\text{residues of } \frac{\beta}{(\lambda - j\beta)(\lambda + j\beta)(1 - e^{-T(s-\lambda)})} \right] \\ &= \frac{\beta}{(\lambda + j\beta)(1 - e^{-T(s-\lambda)})} \Big|_{\lambda = j\beta} + \frac{\beta}{(\lambda - j\beta)(1 - e^{-T(s-\lambda)})} \Big|_{\lambda = -j\beta} \\ &= \frac{1}{2j} \left[\frac{1}{1 - e^{-Ts} e^{j\beta T}} - \frac{1}{1 - e^{-Ts} e^{-j\beta T}} \right] \\ &= \frac{e^{-Ts} \sin \beta T}{1 - 2e^{-Ts} \cos \beta T + e^{-2Ts}} \end{aligned}$$

using the equations from Euler's relation:

$$\cos \beta T = \frac{e^{j\beta T} + e^{-j\beta T}}{2}; \quad \sin \beta T = \frac{e^{j\beta T} - e^{-j\beta T}}{2j}$$

Example 3.5

Example 3.5

Given $e(t) = 1 - e^{-t}$, determine $E^*(s)$, first using (3-3) and then (3-10). Now,

$$E(s) = \frac{1}{s(s+1)}$$

From (3-3),

$$\begin{aligned} E^*(s) &= \sum_{n=0}^{\infty} e(nT) e^{-nTs} \\ &= \sum_{n=0}^{\infty} (1 - e^{-nT}) e^{-nTs} \\ &= \sum_{n=0}^{\infty} e^{-nTs} - \sum_{n=0}^{\infty} e^{-(1+s)nT} \\ &= \frac{1}{1 - e^{-Ts}} - \frac{1}{1 - e^{-(1+s)T}} \end{aligned}$$

From (3-10),

$$\begin{aligned} E^*(s) &= \sum_{\substack{\lambda=0 \\ \lambda=-1}} \left[\text{residues of } \frac{1}{\lambda(\lambda+1)} \frac{1}{1 - e^{-T(s-\lambda)}} \right] \\ &= \frac{1}{1 - e^{-Ts}} - \frac{1}{1 - e^{-(1+s)T}} \end{aligned}$$

Time delay

A delayed signal

$$e(t) = e_1(t - t_0)u(t - t_0)$$

From the shifting property of the Laplace transform (see Appendix VIII),

$$E(s) = \epsilon^{-t_0 s} \mathcal{L}[e_1(t)] = \epsilon^{-t_0 s} E_1(s) \quad (3-12)$$

For this case, (3-10) does not apply; instead, special techniques are required to find the starred transform of a delayed signal in closed form. These techniques will be presented in Chapter 4, where the modified z-transform is developed. However, for the special case in which the time signal is delayed a whole number of sampling periods, (3-10) can be applied in a slightly different form,

$$([\epsilon^{-kTs} E_1(s)])^* = \epsilon^{-kTs} \sum_{\text{at poles of } E_1(\lambda)} \left[\text{residues of } E_1(\lambda) \frac{1}{1 - \epsilon^{-T(s - \lambda)}} \right] \quad (3-13)$$

where k is a positive integer (see Problem 3-7).

See next slide.

Solution of Problem 3-7

$$3-7. \quad E^*(s) = e(0) + e(T)e^{-Ts} + e(2T)e^{-2Ts} + \dots = [e(t)]^*$$

$$\text{Let: } e_1(t) = e(t - kT)u(t - kT)$$

$$\therefore \underline{E_1^*(s) = e(0)e^{-kTs} + e(T)e^{-(k+1)Ts} + \dots}$$

$$= e^{-kTs} [e(0) + e(T)e^{-Ts} + \dots] = e^{-kTs} E^*(s)$$

$\therefore$ from (3-10) ,

$$E_1^*(s) = e^{-kTs} \left[\sum_{\substack{\text{poles} \\ \text{of } E(\lambda)}} \text{residues of } \frac{E(\lambda)}{1 - e^{-T(s-\lambda)}} \right]$$

Example 3.6

Example 3.6

The starred transform of $e(t) = [1 - \epsilon^{-(t-1)}]u(t-1)$, with $T = 0.5$ s, will now be found. First we find $E(s)$:

$$E(s) = \frac{\epsilon^{-s}}{s} - \frac{\epsilon^{-s}}{s+1} = \frac{\epsilon^{-s}}{s(s+1)}$$

Therefore, in (3-13), k is equal to 2 and

$$E_1(s) = \frac{1}{s(s+1)}$$

Then, from (3-13),

$$\begin{aligned} \left[\frac{\epsilon^{-s}}{s(s+1)} \right]^* &= \epsilon^{-s} \sum_{\lambda=0, -1} \left[\text{residues of } \frac{1}{\lambda(\lambda+1)(1 - \epsilon^{-0.5(s-\lambda)})} \right] \\ &= \epsilon^{-s} \left[\frac{1}{(\lambda+1)(1 - \epsilon^{-0.5(s-\lambda)})} \Big|_{\lambda=0} + \frac{1}{\lambda(1 - \epsilon^{-0.5(s-\lambda)})} \Big|_{\lambda=-1} \right] \\ &= \epsilon^{-s} \left[\frac{1}{1 - \epsilon^{-0.5s}} + \frac{-1}{1 - \epsilon^{-0.5(s+1)}} \right] = \frac{(1 - \epsilon^{-0.5})\epsilon^{-1.5s}}{(1 - \epsilon^{-0.5s})(1 - \epsilon^{-0.5(s+1)})} \end{aligned}$$

3.5 Results From the Fourier Transform

In this section we present some results regarding the Fourier transform. These results are helpful in understanding the effects of sampling a signal.

The Fourier transform is defined by [4]

$$\mathcal{F}[e(t)] = E(j\omega) = \int_{-\infty}^{\infty} e(t)\epsilon^{-j\omega t} dt \quad (3-14)$$

In many texts the notation $E(\omega)$ is used instead of $E(j\omega)$; we discuss this further below.

$$\mathcal{L}[e(t)] = E(s) = \int_0^{\infty} e(t)\epsilon^{-st} dt \quad (3-15)$$

If $e(t)$ is zero for $t < 0$, its Fourier transform is given by

$$\begin{aligned} \mathcal{F}[e(t)] &= E(j\omega) = \int_{-\infty}^0 e(t)\epsilon^{-j\omega t} dt + \int_0^{\infty} e(t)\epsilon^{-j\omega t} dt \\ &= \int_0^{\infty} e(t)\epsilon^{-j\omega t} dt = \mathcal{L}[e(t)] \Big|_{s=j\omega} \end{aligned}$$

provided that both transforms exists. This result can be expressed in general as

$$\mathcal{F}[e(t)u(t)] = \mathcal{L}[e(t)u(t)] \Big|_{s=j\omega} \quad (3-16)$$

Hence, for the case that $e(t)$ is zero for negative time, the Fourier transform of $e(t)$ is equal to the Laplace transform of $e(t)$ with s replaced with $j\omega$. We see then the reason for expressing the Fourier transform as $E(j\omega)$. In (3-16) it is assumed that both the Fourier transform and the Laplace transform exist. Recall that neither transform may exist for a given time function, or that the Laplace transform may exist while the Fourier transform does not.

A plot of the Fourier transform $E(j\omega)$ is called the frequency spectrum of $e(t)$. A common procedure for showing the frequency spectrum is to express $E(j\omega)$ as

$$E(j\omega) = |E(j\omega)|e^{j\theta(j\omega)} = |E(j\omega)| \angle \theta(j\omega)$$

and plot $|E(j\omega)|$ versus ω (the amplitude spectrum) and $\theta(j\omega)$ versus ω (the phase spectrum). We interpret the amplitude spectrum as giving the relative contents of a signal in different frequency bands. The meaning of the frequency spectrum of a signal is fundamental to understanding the transmission of signals through systems; those readers who do not have a good background in this area should review this material in a signals and systems book such as Ref. 4.

System TF: $G(s)$

Consider now an analog system with the input signal $e(t)$, the output signal $y(t)$, and the transfer function $G(s)$; that is,

$$Y(s) = \underline{G(s)}E(s) \quad (3-17)$$

If $e(t)$ is the unit impulse function $\delta(t)$, $E(s) = 1$ and $Y(s) = G(s)$. Hence $g(t) = \mathcal{L}^{-1}[G(s)]$ is the unit impulse response of the system with the transfer function $G(s)$.

A physical system cannot respond to an input signal before this signal is applied. For this case, $g(t) = 0$ for $t < 0$. A system that satisfies this condition is called a causal system. Consequently, for a causal system, the Fourier transform of the unit impulse response is equal to the Laplace transform of that response with s replaced with $j\omega$, provided that the Fourier transform exists. Hence, for the system of (3-17), the Fourier transform yields

$$Y(j\omega) = \underline{G(j\omega)}E(j\omega)$$

We see that $G(j\omega)$ determines the manner that the frequency spectrum of the input $e(t)$ is modified to yield the frequency spectrum of the output $y(t)$. The function $G(j\omega)$ is called the frequency response.

3.6 Properties of $E^*(s)$

Two s -plane properties of $E^*(s)$ are now given. These properties are very important and will be used extensively in later derivations.

Property 1. $E^*(s)$ is periodic in s with period $j\omega_s$

This property can be proved using either (3-3), (3-10), or (3-11). From (3-3),

$$E^*(s + jm\omega_s) = \sum_{n=0}^{\infty} e(nT) e^{-nT(s + jm\omega_s)} \quad (3-18)$$

Since $\omega_s T = (2\pi/T)T = 2\pi$, and from Euler's relationship,

$$e^{j\theta} = \cos \theta + j \sin \theta$$

then

$$e^{-jnm\omega_s T} = e^{-jnm2\pi} = 1, \quad m \text{ an integer} \quad (3-19)$$

Thus, in (3-18),

$$\underline{E^*(s + jm\omega_s)} = \sum_{n=0}^{\infty} e(nT) e^{-nTs} = \underline{E^*(s)} \quad (3-20)$$

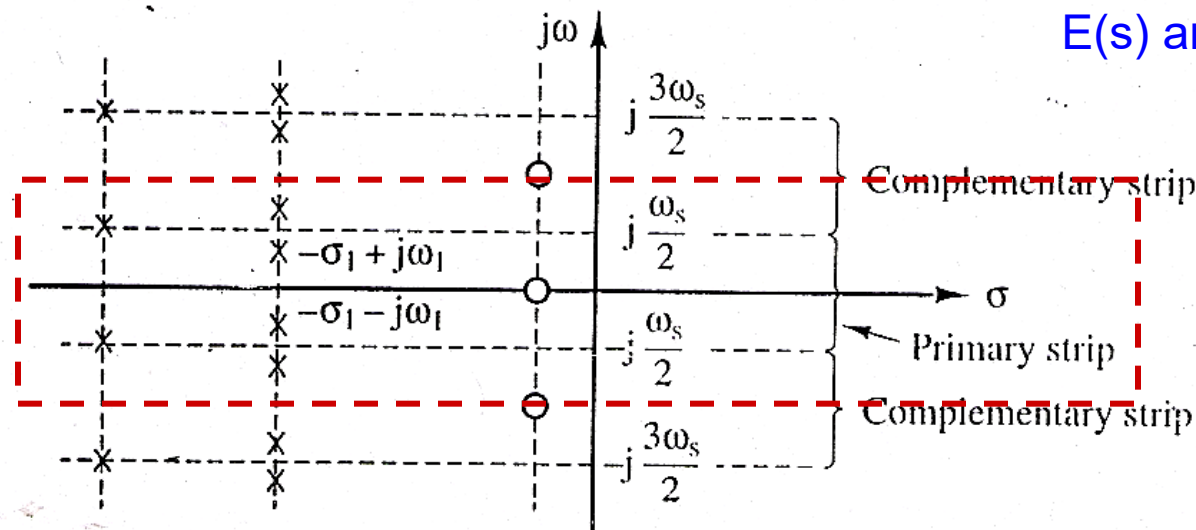
Property 2. If $E(s)$ has a pole at $s = s_1$, then $E^*(s)$ must have poles at $s = s_1 + jm\omega_s, m = 0, \pm 1, \pm 2, \dots$

This property can be proved from (3-11). Consider $e(t)$ to be continuous at all sampling instants. Then

$$E^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} E(s + jn\omega_s) = \frac{1}{T} [E(s) + E(s + j\omega_s) + E(s + 2j\omega_s) + \dots + E(s - j\omega_s) + E(s - j2\omega_s) + \dots] \quad (3-21)$$

If $E(s)$ has a pole at $s = s_1$, then each term of (3-21) will contribute a pole at $s = s_1 + jm\omega_s$, where m is an integer.

How about the zero locations of $E(s)$ and $E^*(s)$?



Primary Strip

Figure 3-8 Pole-zero locations for $E^*(s)$.

$$\omega_1 = \frac{\pi/2}{T}, \because \omega_s = \frac{2\pi}{T}$$

$$\Rightarrow \omega_s = 4\omega_1$$

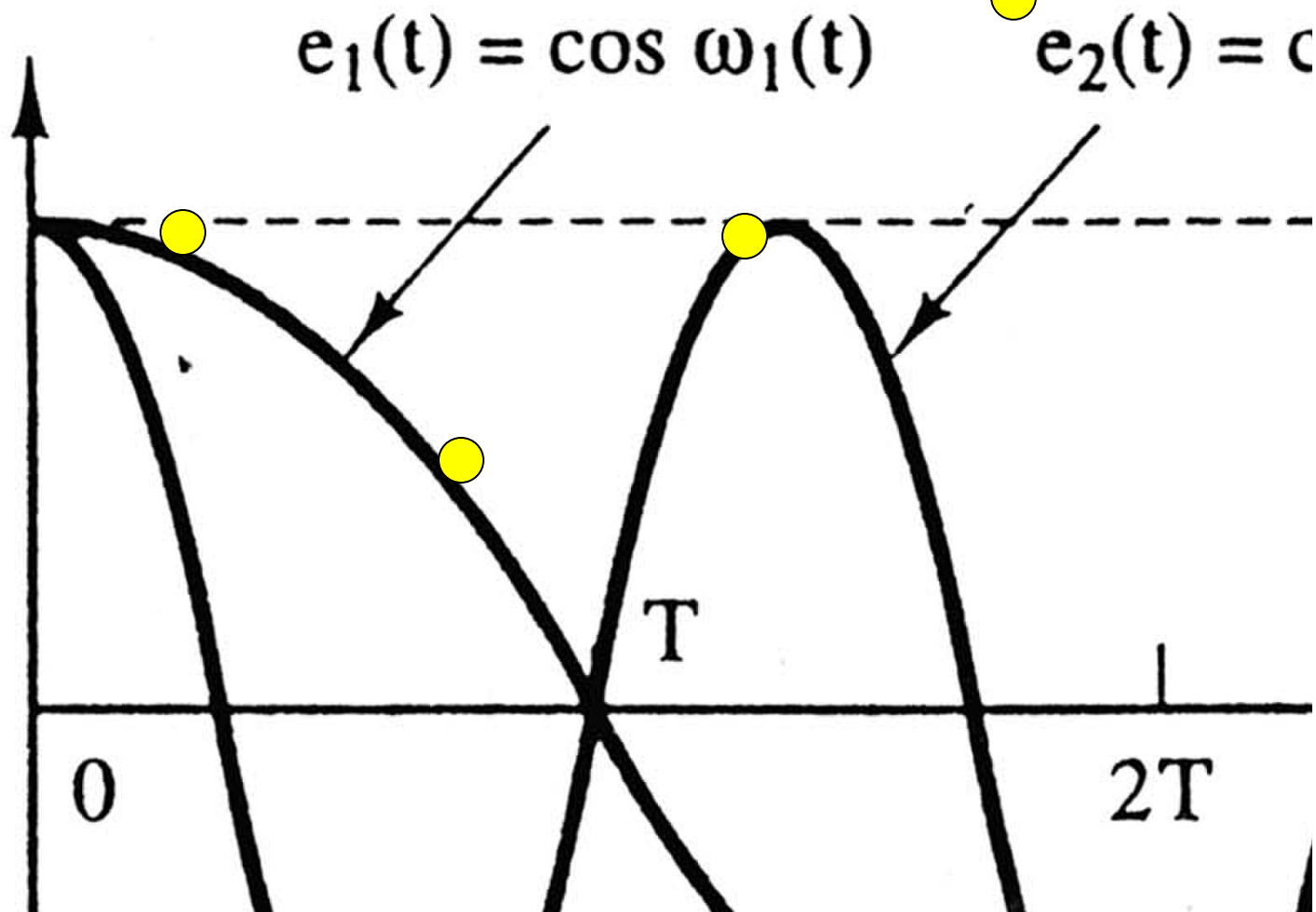
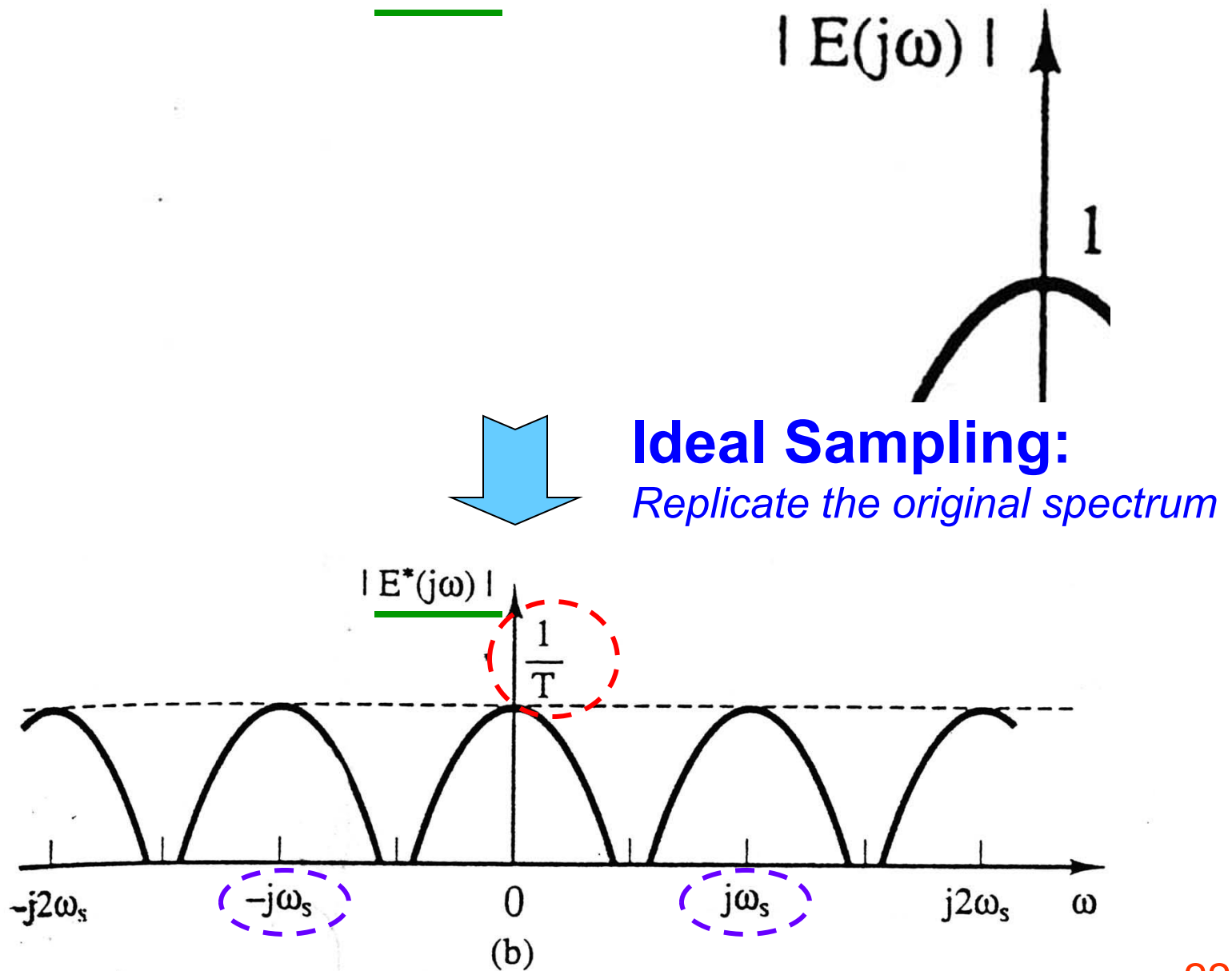


Fig. 3-10 Freq. Spectra



Shannon's Sampling Theorem (1949)

- If a function $e(t)$ contains **no frequencies higher than f_0 hertz**, it is completely determined by giving its ordinates at a series of points spaced **$1/(2f_0)$ seconds** apart.
 - $e(t)$ is **band-limited** to one-sided baseband bandwidth f_0
 - f_s : **sampling rate**

$$f_s > 2f_0$$

Nyquist rate: $2f_0$
Nyquist freq: $f_s/2$

- The theorem is commonly called the **Nyquist sampling theorem**.
- But, why Nyquist?

Aliasing (Freq. fold-over)

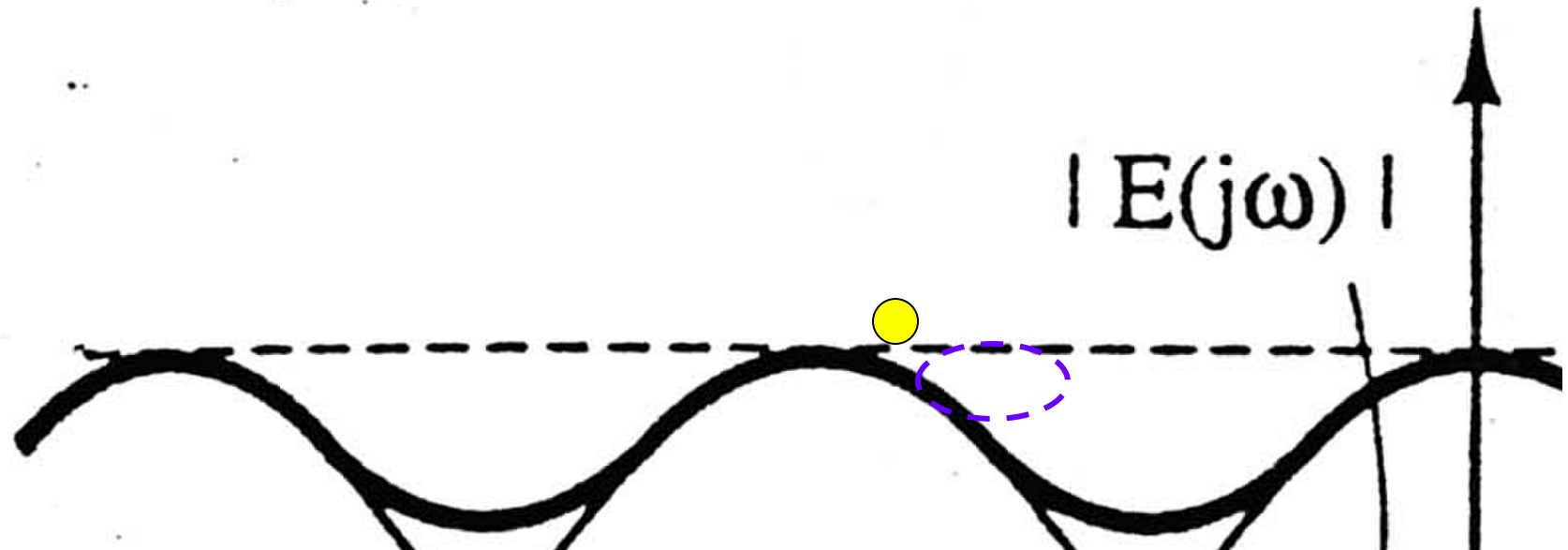


Figure 3-10 Freq

Ideal Filter (P101):

- *Unity Gain: passband*
- *Zero Gain: outside the passband*

3.7 Data Reconstruction

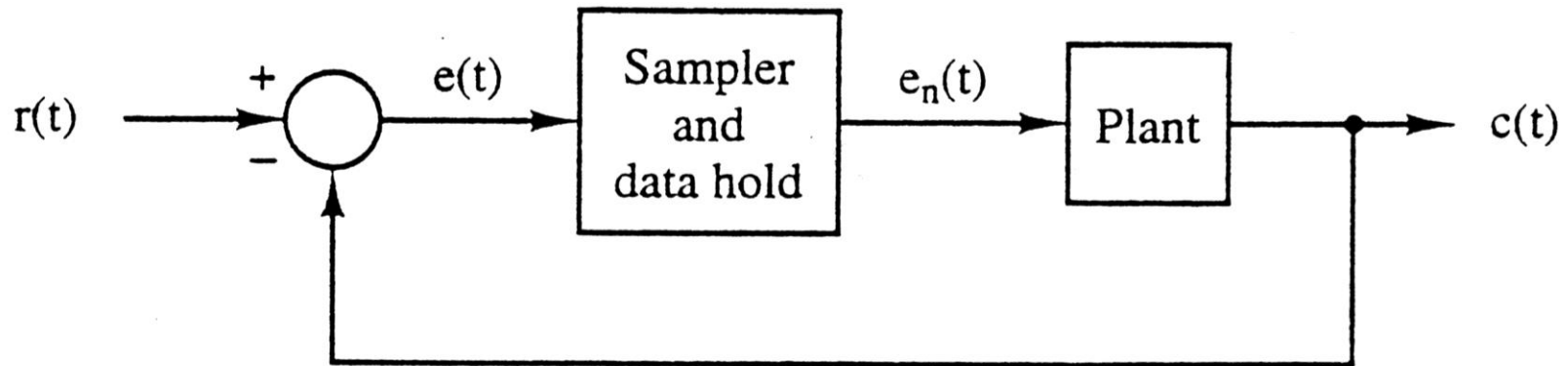


Figure 3-11 Sampled-data control system.

- $e(t)$ is **band-limited** in freq.
- **Reconstruction**
 - Theoretical: by **an ideal low pass filter**.
 - Practical: by approximations.
 - e.g. **Taylor's series expansion** about $t = nT$.

Taylor's series

$$e(t) = e(nT) + e'(nT)(t - nT) + \frac{e''(nT)}{2!}(t - nT)^2 + \dots \quad (3-23)$$

where the prime denotes the derivative. $e_n(t)$ is defined as the reconstructed version of $e(t)$ for the n th sample period; that is,

$$e_n(t) \cong e(t) \quad \text{for } nT \leq t < (n+1)T \quad (3-24)$$

Thus $e_n(t)$ denotes the output of the data hold. Since $e(t)$ enters the data hold only in sampled form, the values of the derivatives are not known. However, the derivatives may be approximated by the backward difference

$$e'(nT) = \frac{1}{T}[e(nT) - e[(n-1)T]] \quad (3-25)$$

$$e''(nT) = \frac{1}{T}[e'(nT) - e'[(n-1)T]] \quad (3-26)$$

and so on. Note that by substituting (3-25) into (3-26), we obtain

$$e''(nT) = \frac{1}{T} \left[\frac{1}{T}[e(nT) - e[(n-1)T]] - \frac{1}{T}[e[(n-1)T] - e[(n-2)T]] \right]$$

or

$$e''(nT) = \frac{1}{T^2}[e(nT) - 2e[(n-1)T] + e[(n-2)T]] \quad (3-27)$$

Three types of data hold based on the foregoing relationships—the zero-, first-, and fractional-order holds—will now be discussed.

Zero-Order Hold (ZOH)

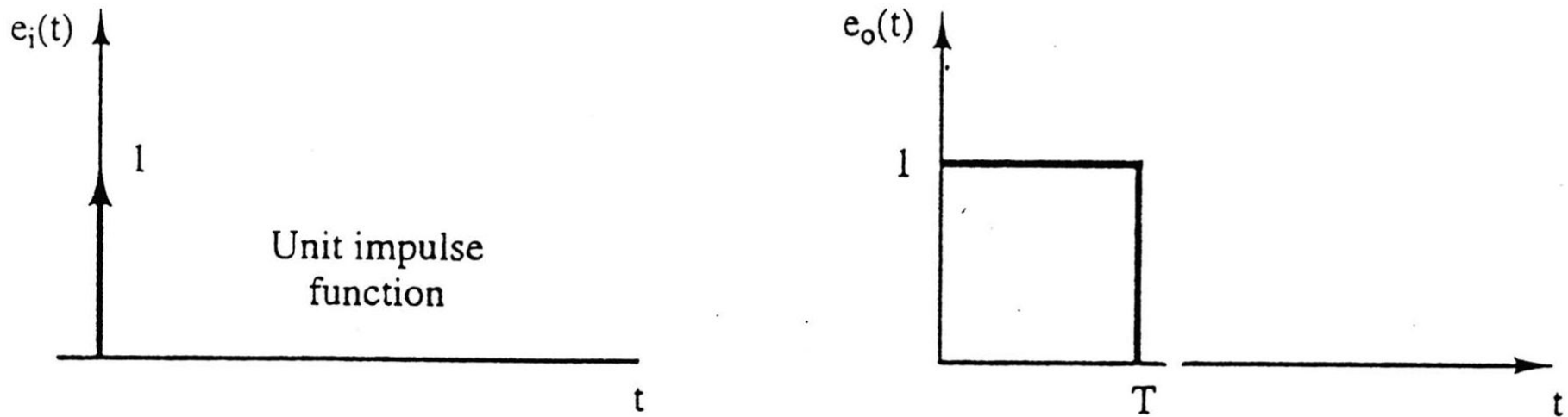


Figure 3-12 Input and output signals for the zero-order hold.

$$e_n(t) = e(nT), \quad nT \leq t < (n+1)T \quad (3.28) \quad \text{cf. (3.1)}$$

$$e_o(t) = u(t) - u(t - T)$$

$$\Rightarrow G_{h0}(s) = \frac{E_o(s)}{E_i(s)} = \frac{1 - e^{-Ts}}{s} \quad (3.29)$$

Frequency Resp. of ZOH

$s = j\omega$

$$\begin{aligned} G_{h0}(j\omega) &= \frac{1 - e^{-j\omega T}}{j\omega} e^{j(\omega T/2)} e^{-j(\omega T/2)} = \frac{2e^{-j(\omega T/2)}}{\omega} \left[\frac{e^{j(\omega T/2)} - e^{-j(\omega T/2)}}{2j} \right] \\ &= T \frac{\sin(\omega T/2)}{\omega T/2} e^{-j(\omega T/2)} \end{aligned} \quad (3-30)$$

Since

$$\frac{\omega T}{2} = \frac{\omega}{2} \left(\frac{2\pi}{\omega_s} \right) = \frac{\pi\omega}{\omega_s}$$

(3-30) can be expressed as

$$G_{h0}(j\omega) = T \frac{\sin(\pi\omega/\omega_s)}{\pi\omega/\omega_s} e^{-j(\pi\omega/\omega_s)} \quad (3-31)$$

Magnitude & Phase

$$|G_{h0}(j\omega)| = T \left| \frac{\sin(\pi\omega/\omega_s)}{\pi\omega/\omega_s} \right| \quad (3-32)$$

and

$$\angle G_{h0}(j\omega) = -\frac{\pi\omega}{\omega_s} + \epsilon, \quad \theta = \begin{cases} 0, & \sin\left(\frac{\pi\omega}{\omega_s}\right) > 0 \\ \pi, & \sin\left(\frac{\pi\omega}{\omega_s}\right) < 0 \end{cases} \quad (3-33)$$

The amplitude and phase plots for $G_{h0}(j\omega)$ are shown in Figure 3-13.

Frequency Resp. of ZOH

Approximation of an ideal LP filter.

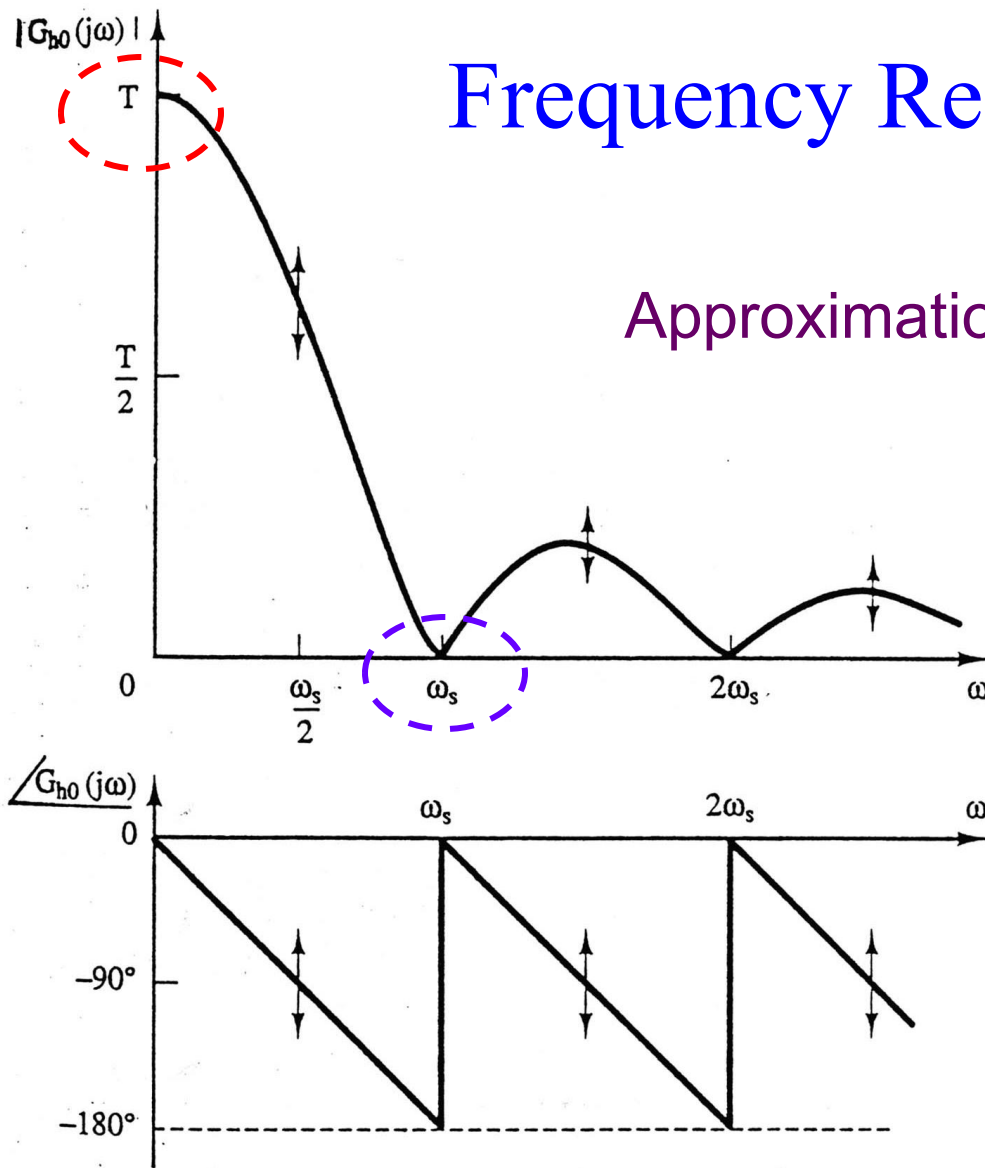
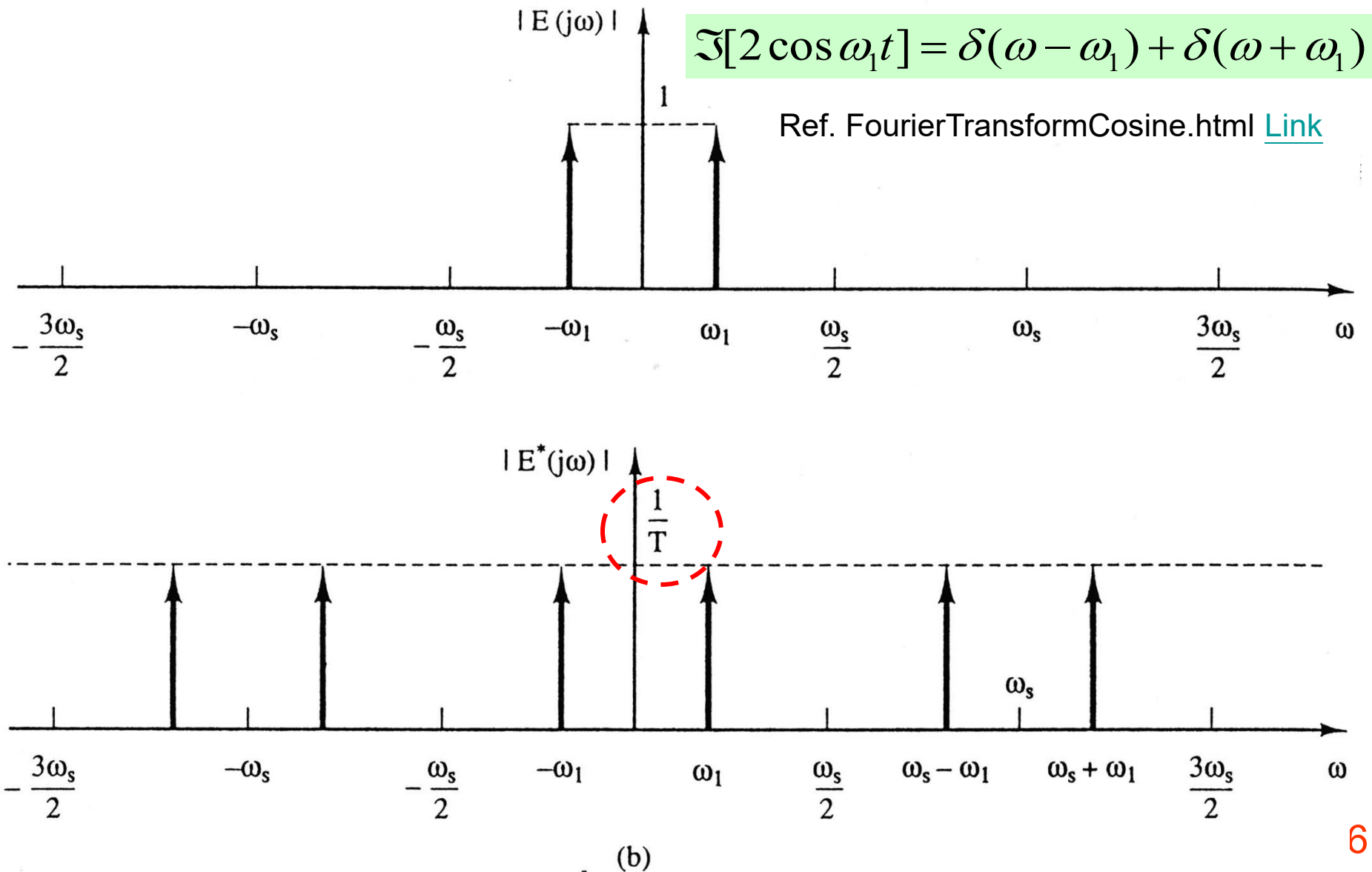


Figure 3-13 Frequency response of the zero-order hold.

Example: $e(t) = 2 \cos \omega_1 t$



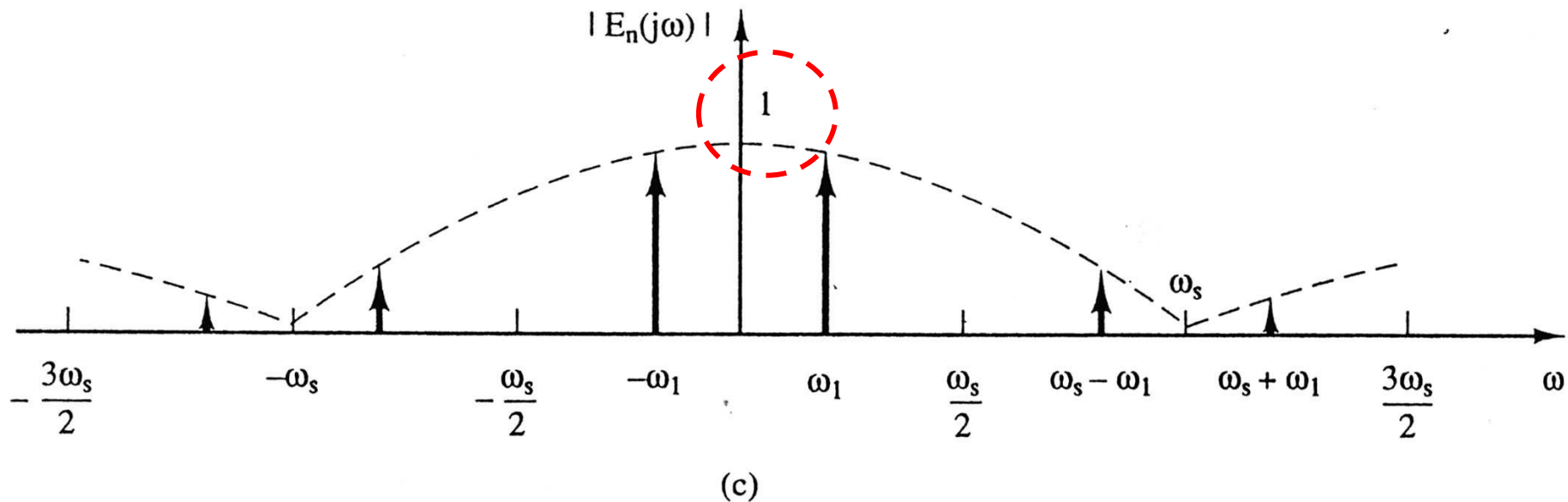


Figure 3-14 Sinusoidal response of the sampler and zero-order hold; (a) input signal to the ideal sampler; (b) output signal of the ideal sampler; (c) output signal from the zero-order hold.

Compare the I/O signals in time and frequency domains.

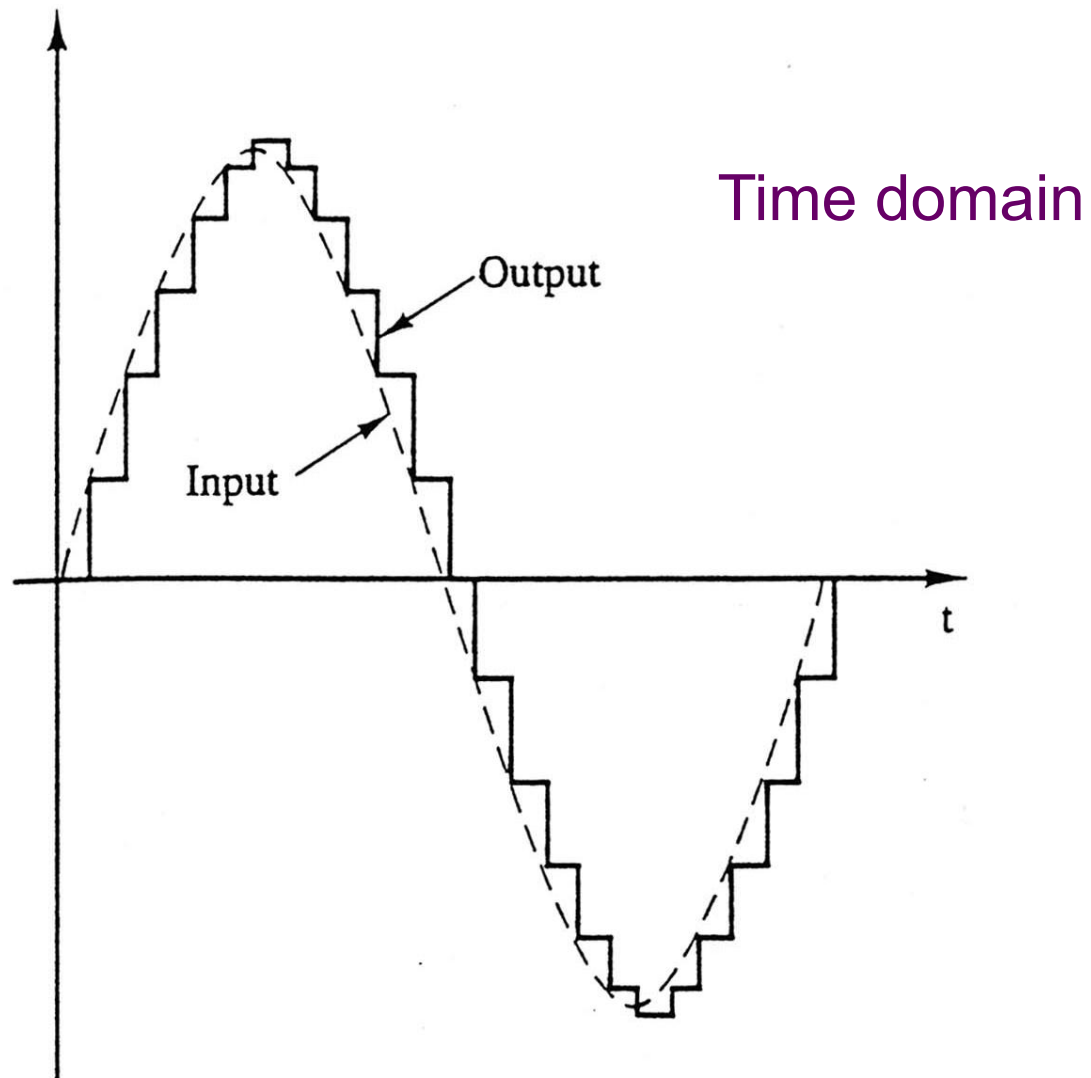


Figure 3-15 Output of a sampler/zero-order hold for a sinusoidal input.

First-Order Hold

First-Order Hold

The first two terms of (3-23) are used to realize the first-order hold. Therefore,

$$e_n(t) = e(nT) + e'(nT)(t - nT), \quad nT \leq t < (n + 1)T \quad (3-34)$$

where, from (3-25),

$$\underline{e'(nT) = \frac{e(nT) - e[(n - 1)T]}{T}} \quad (3-35)$$

To determine the transfer function of a first-order hold, assume that the input is a unit impulse function. Then, from (3-34) and (3-35), the output, shown in Figure 3-16a, is

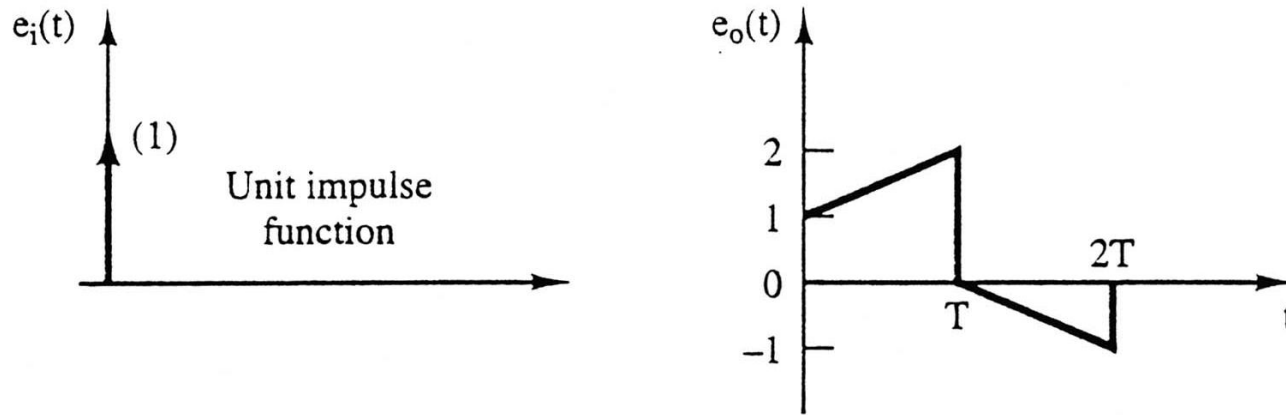
$$\begin{aligned} e_o(t) = & u(t) + \frac{1}{T}tu(t) - 2u(t - T) - \frac{2}{T}(t - T)u(t - T) + u(t - 2T) \\ & + \frac{1}{T}(t - 2T)u(t - 2T) \end{aligned}$$

Since $E_i(s) = 1$,

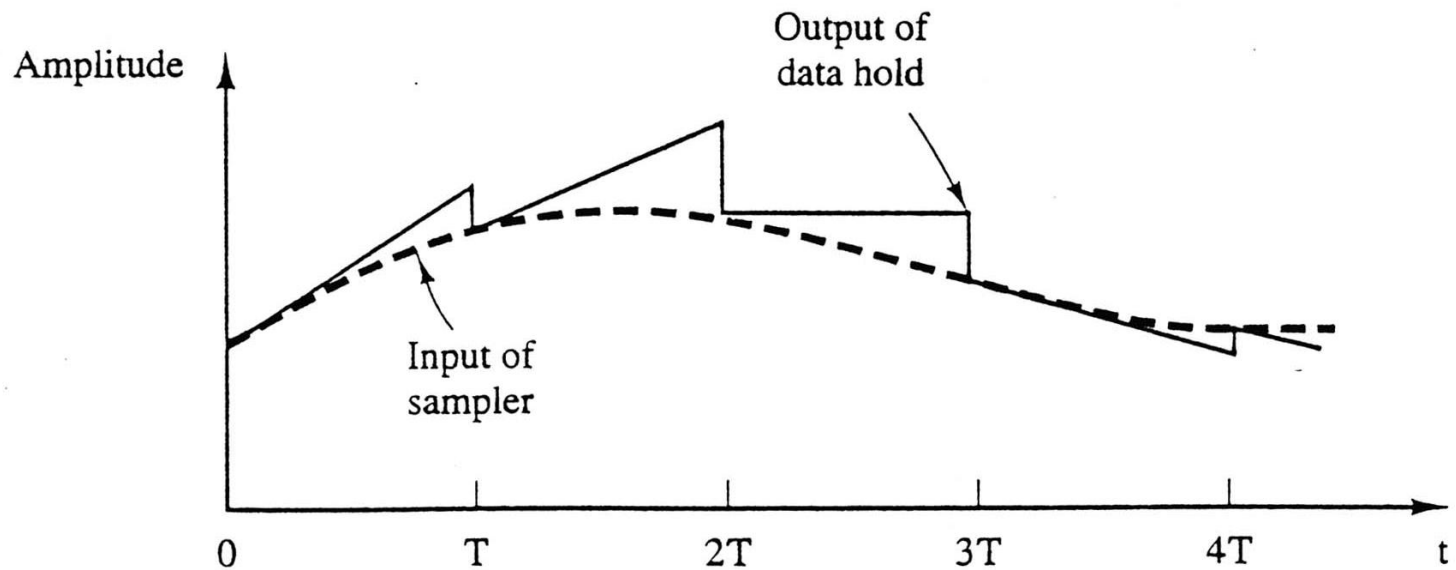
$$G_{h1}(s) = \frac{E_o(s)}{E_i(s)} = \frac{1}{s} - \frac{2\epsilon^{-Ts}}{s} + \frac{\epsilon^{-2Ts}}{s} + \frac{1}{Ts^2}(1 - 2\epsilon^{-Ts} + \epsilon^{-2Ts})$$

or

$$G_{h1}(s) = \frac{1 + Ts}{T} \left[\frac{1 - \epsilon^{-Ts}}{s} \right]^2$$



(a)



(b)

Figure 3-16 Example of the output of a first-order hold.

An example of the input–output waveforms of a sampler and first-order hold is shown in Figure 3-16b.

The frequency response of the first-order hold is obtained from (3-36).

$$G_{h1}(j\omega) = \frac{1 + j\omega T}{T} \left[\frac{1 - e^{-j\omega T}}{j\omega} \right]^2$$

Magnitude & Phase

$$|G_{h1}(j\omega)| = T \sqrt{1 + \frac{4\pi^2 \omega^2}{\omega_s^2} \left[\frac{\sin(\pi\omega/\omega_s)}{\pi\omega/\omega_s} \right]^2} \quad (3-37)$$

$$\angle G_{h1}(j\omega) = \tan^{-1}\left(\frac{2\pi\omega}{\omega_s}\right) - \frac{2\pi\omega}{\omega_s} \quad (3-38)$$

Frequency Resp. of 1st-OH

Approximation of an ideal LP filter.

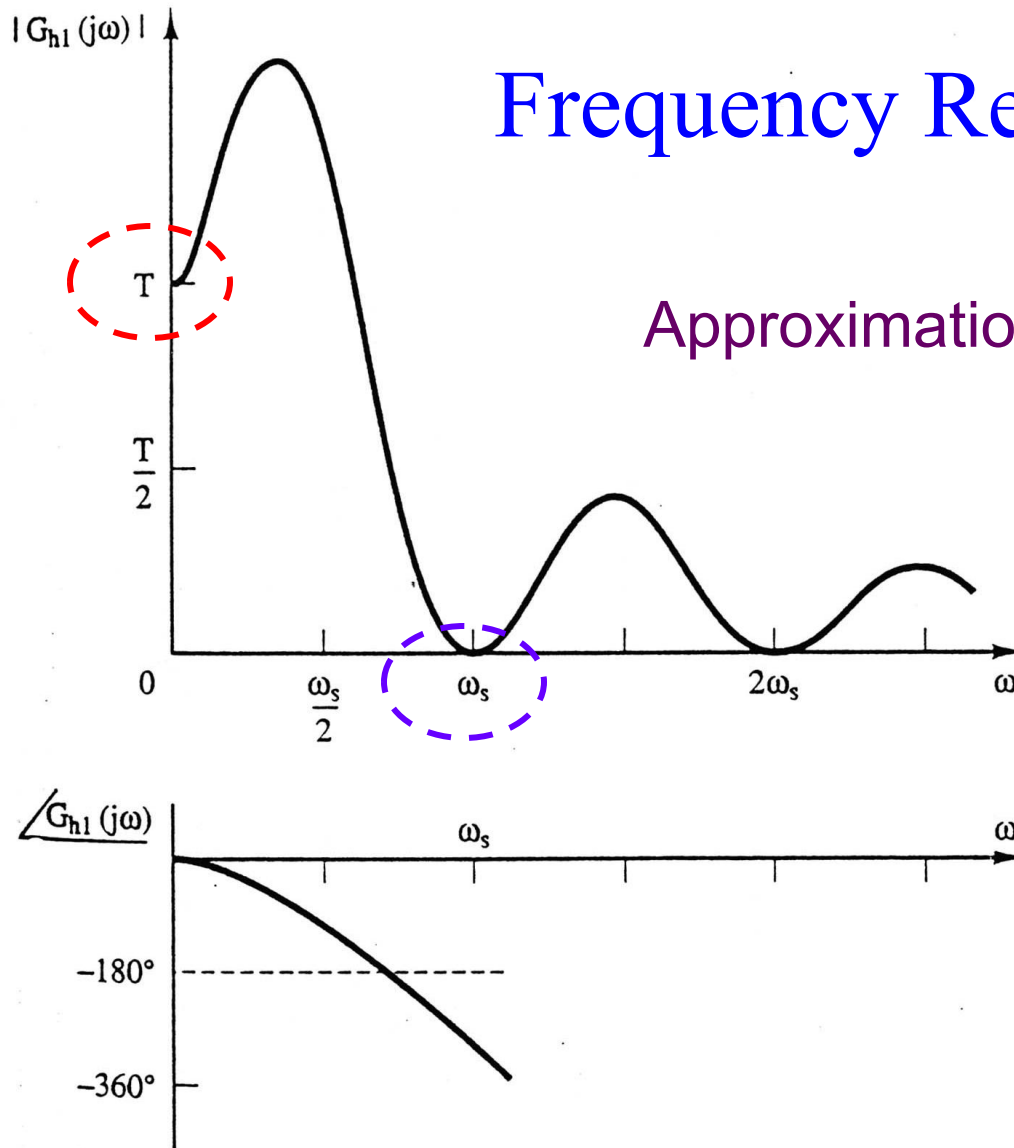
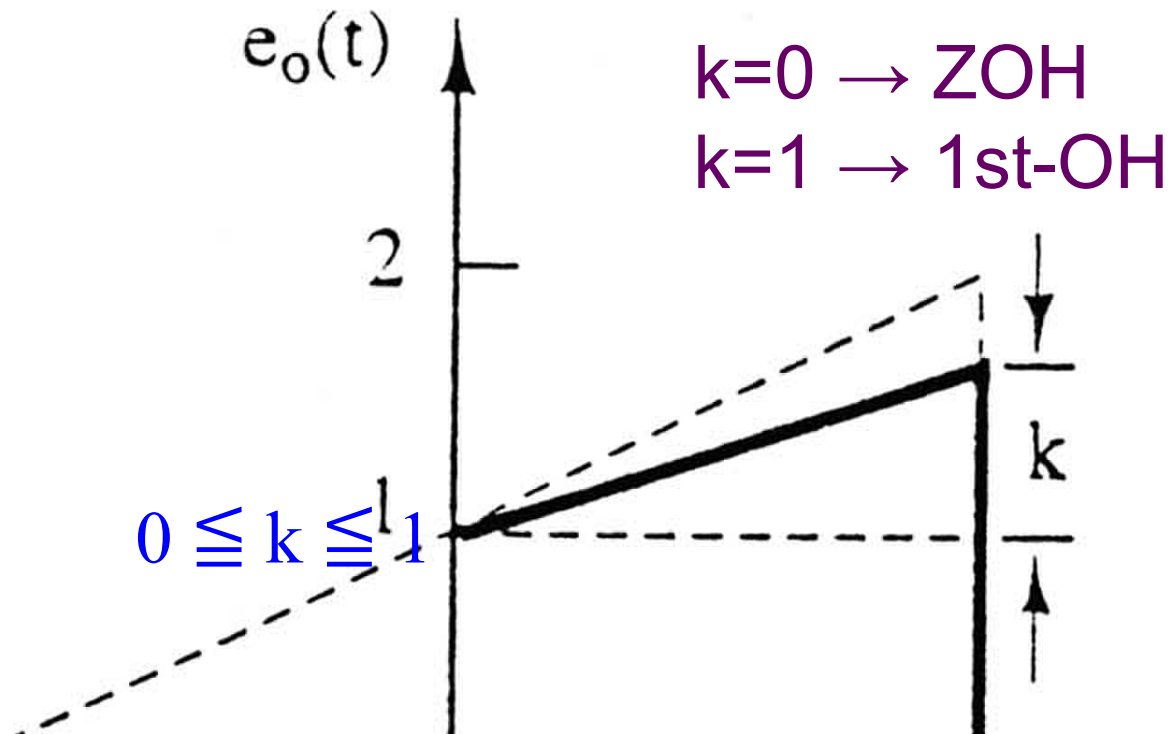
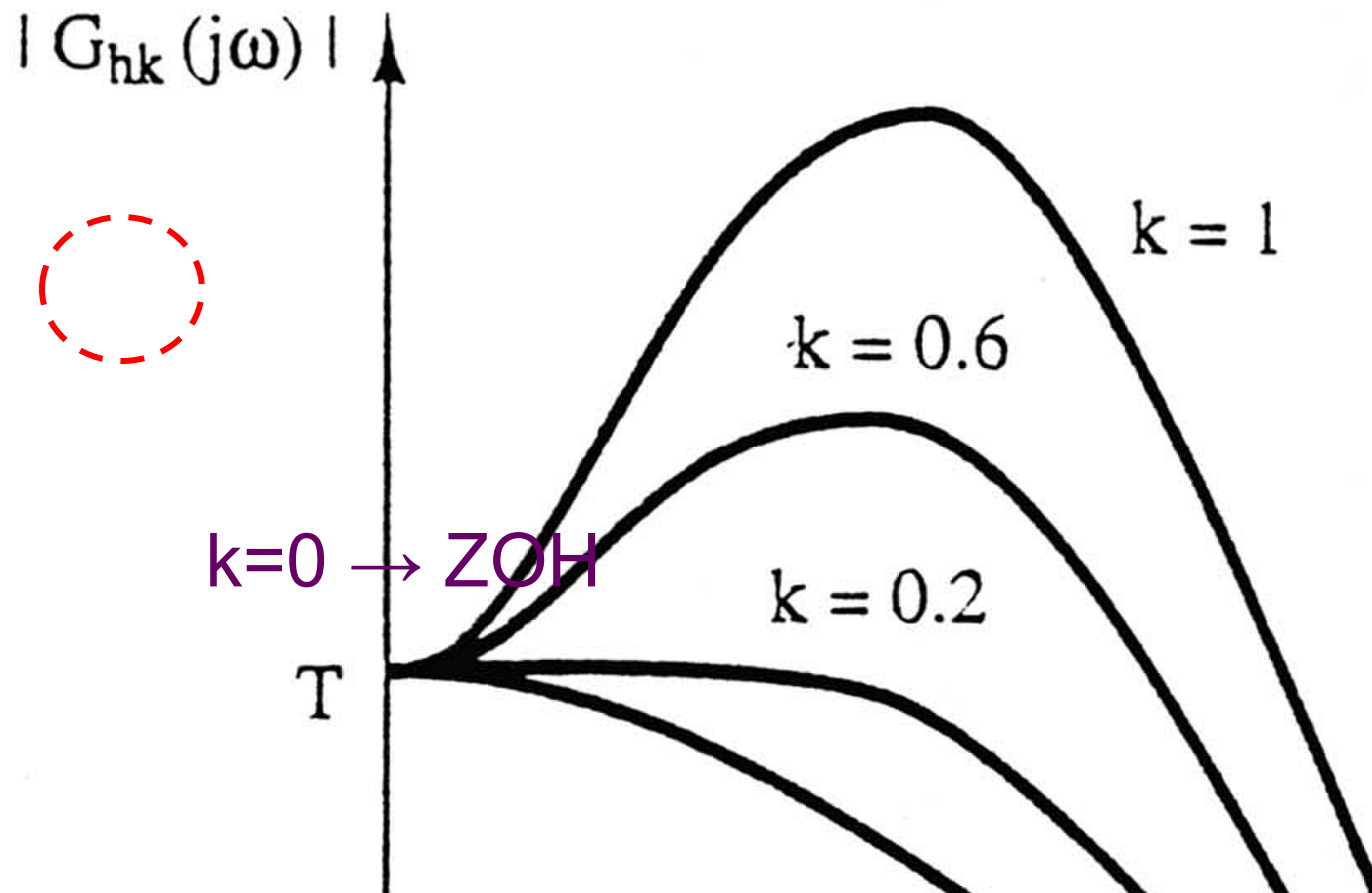


Figure 3-17 Frequency response of the first-order hold.

Fractional-Order Hold



$k=1 \rightarrow 1\text{st-OH}$



Supplementary Data for Ch3.

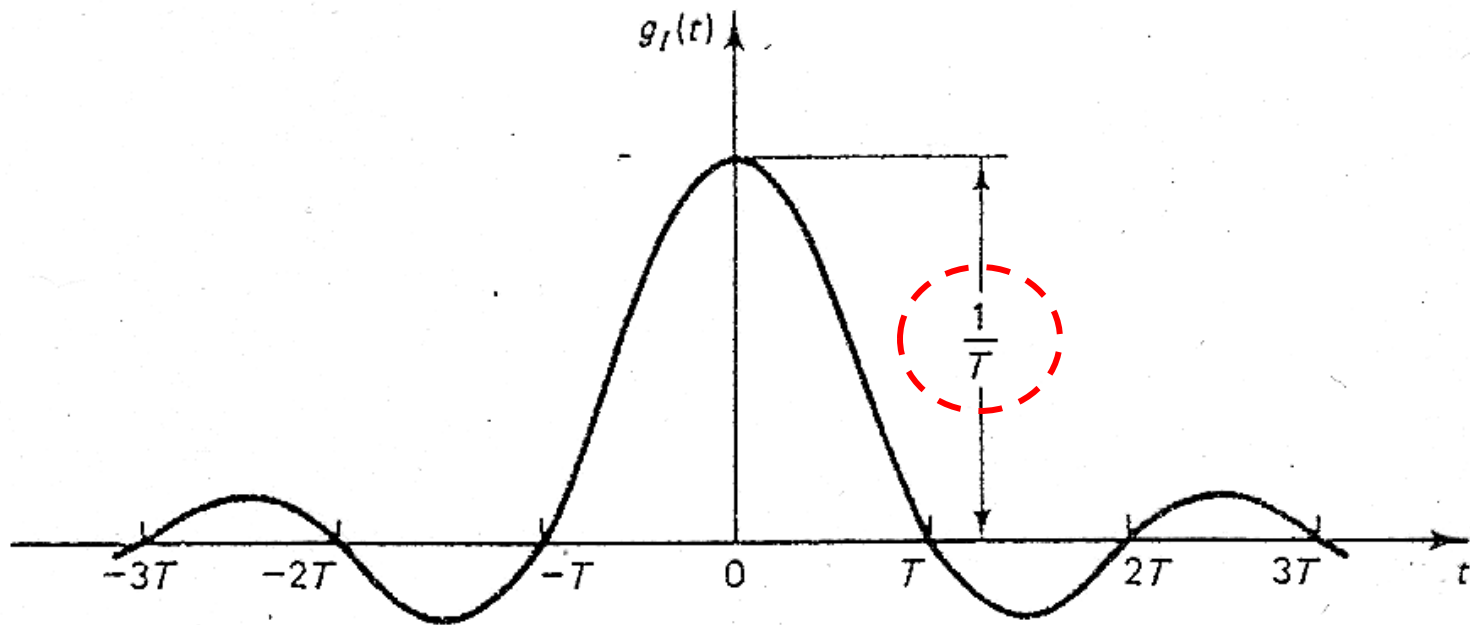


Figure 3-14 Impulse response $g_I(t)$ of ideal filter.

Ideal Filter (P101):

- *Unity Gain: passband*
- *Zero Gain: outside the passband*

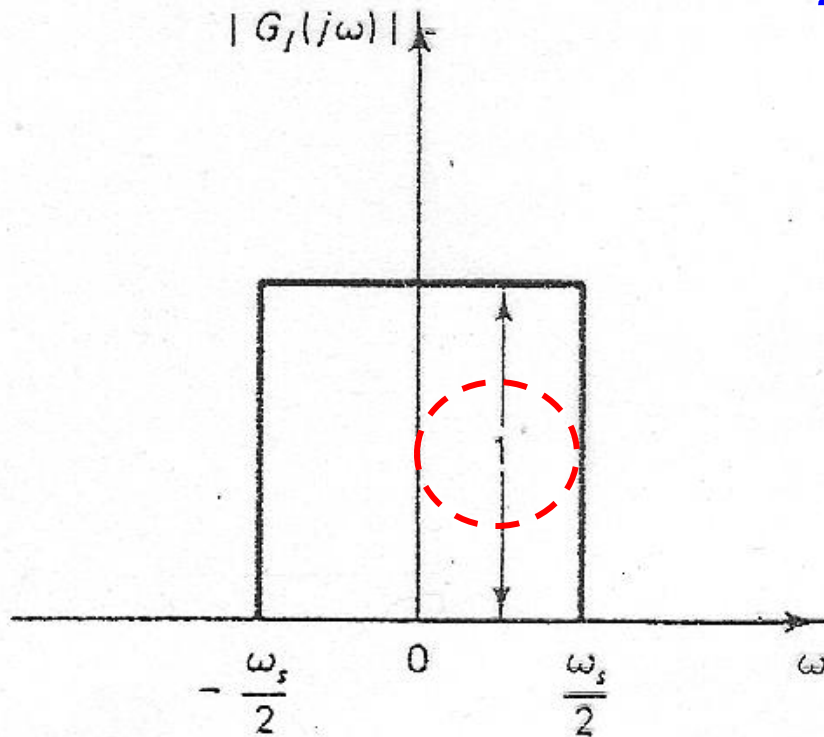


Figure 3-12 Amplitude frequency spectrum of the ideal low-pass filter.

Sampling & Ideal Filtering

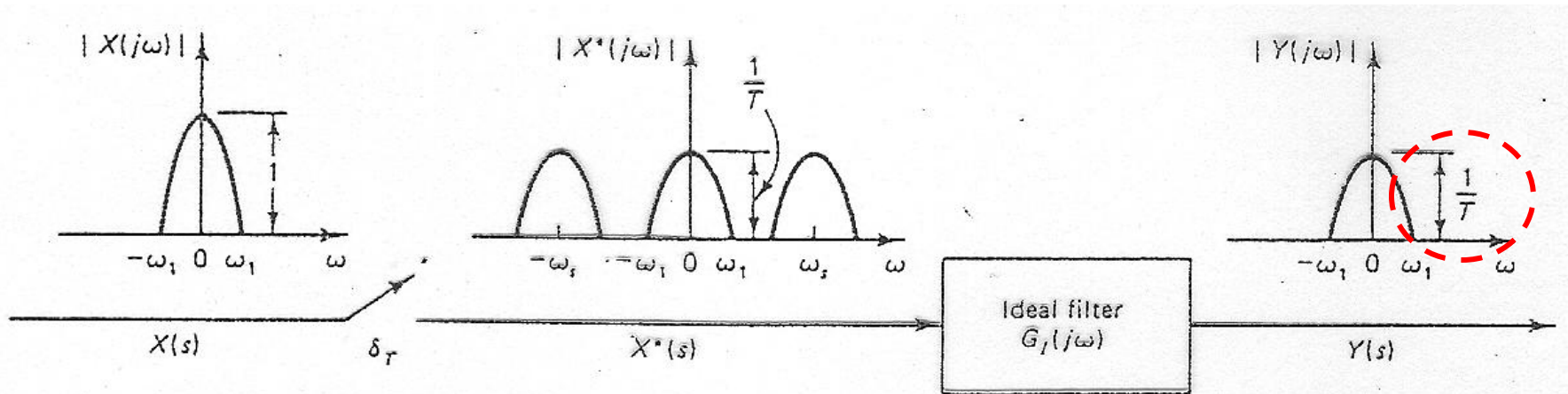


Figure 3-13 Frequency spectra of the signals before and after ideal filtering.

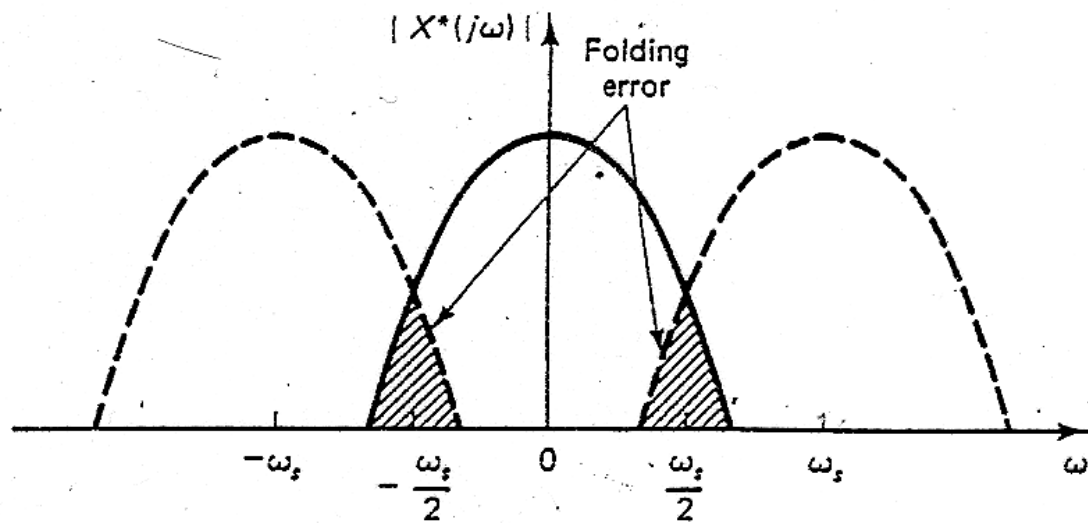


Figure 3-17 Diagram showing the regions where folding error occurs.

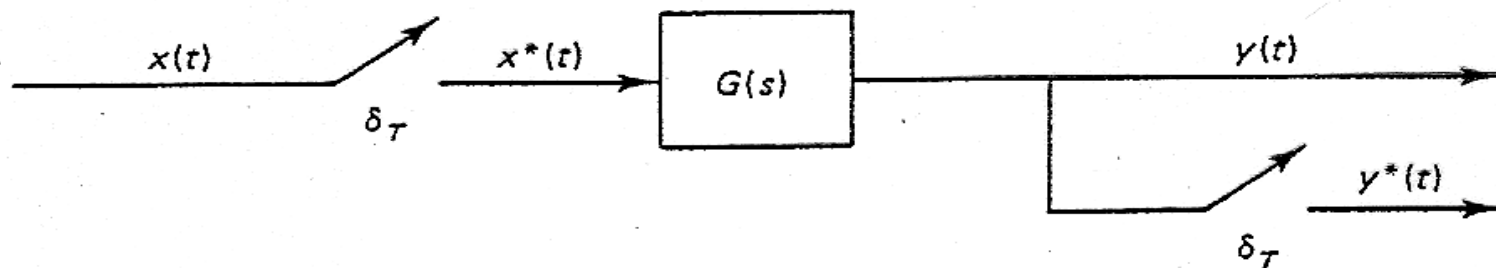
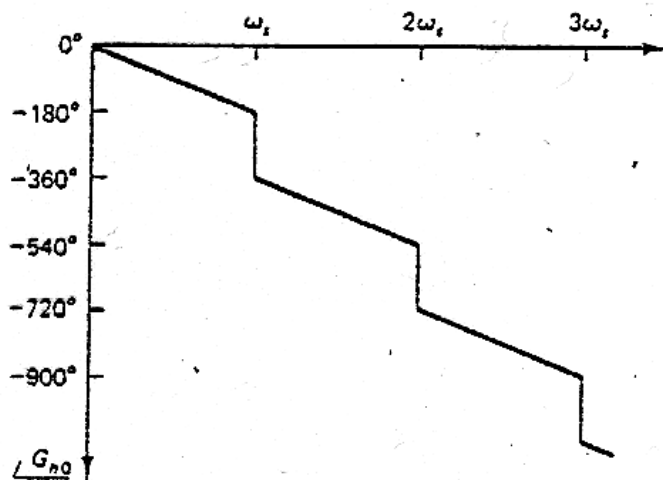
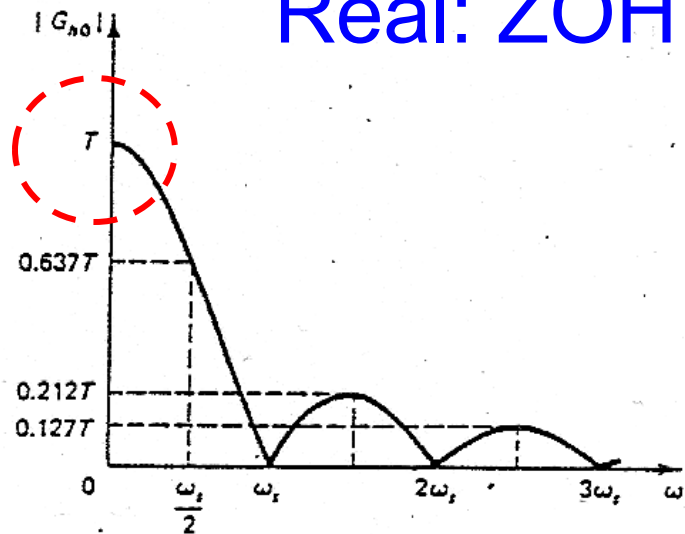


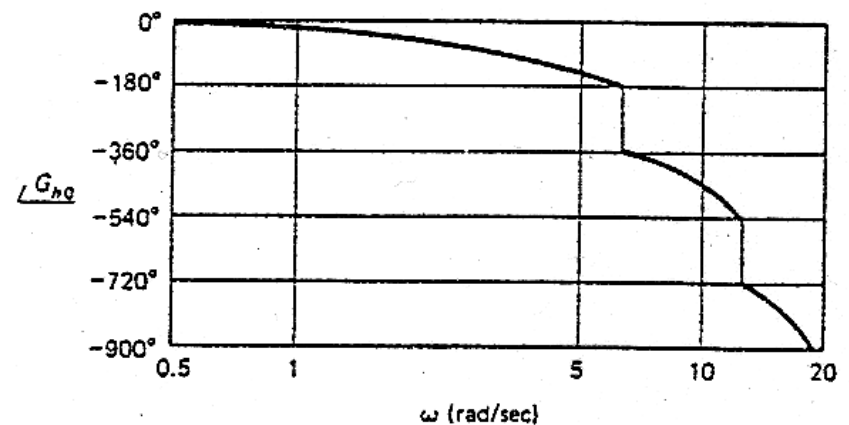
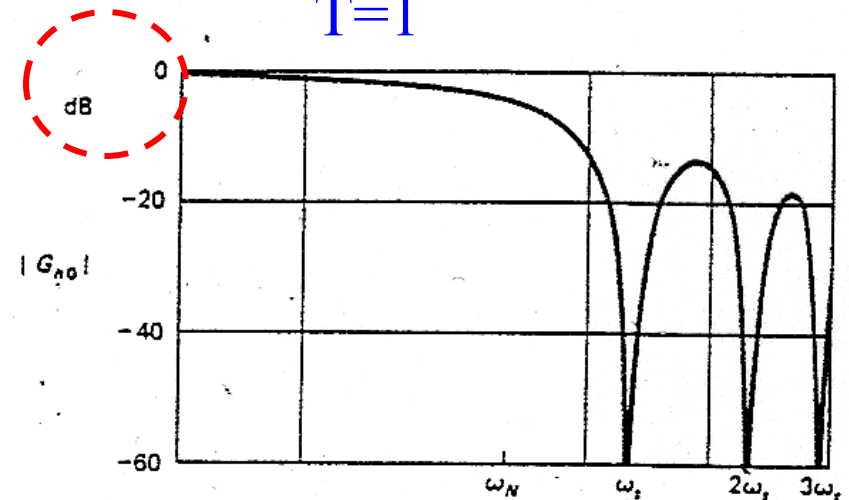
Figure 3-20 Continuous-time system $G(s)$ driven by an impulse-sampled signal.

Real: ZOH



(a)

$T=1$



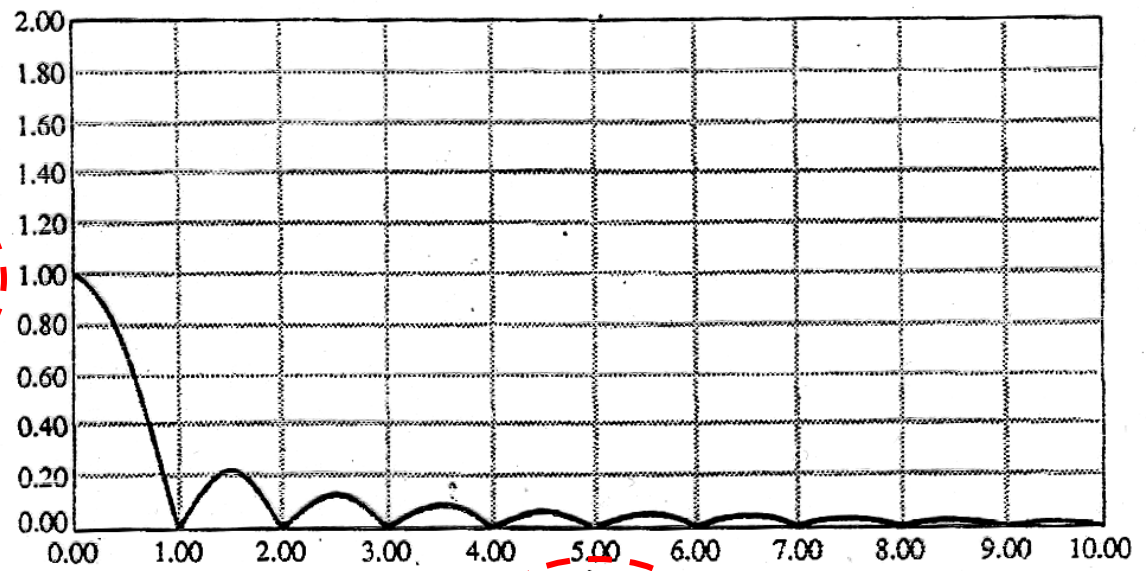
(b)

Figure 3-15 (a) Frequency-response curves for the zero-order hold; (b) equivalent Bode diagram when $T = 1$ sec.

Real: ZOH

Normalized

$|G_{HO}(j\omega)|/T$



Scaled

ω/ω_s

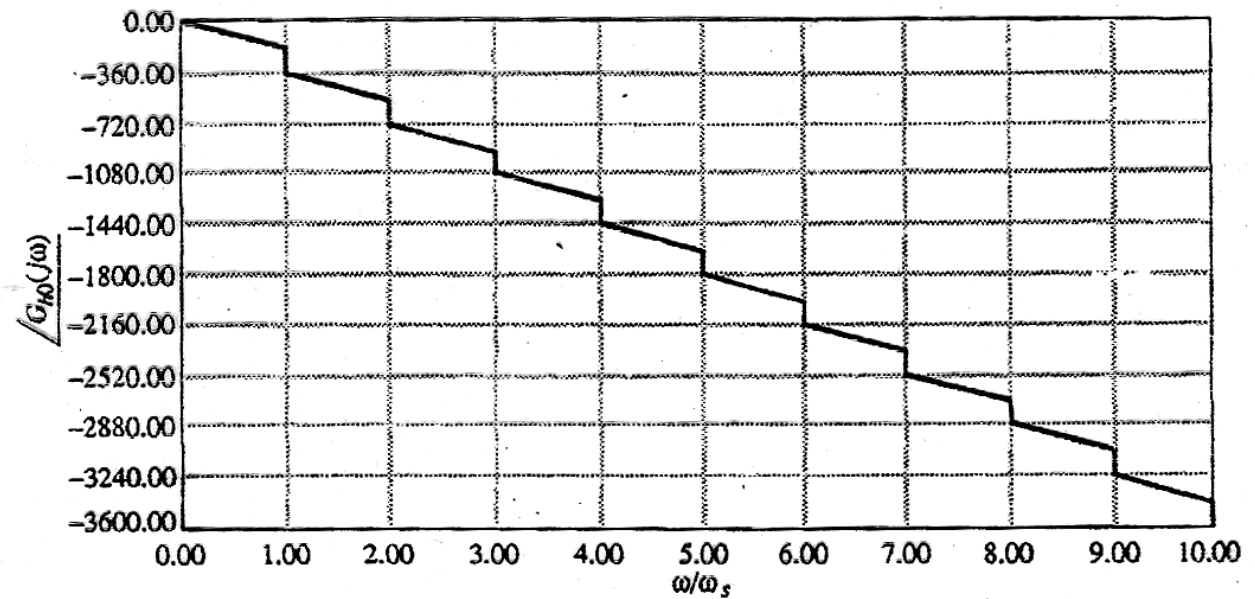


Figure 2-35. Gain and phase characteristics of the zero-order hold.

Ideal Filter vs. Real ZOH

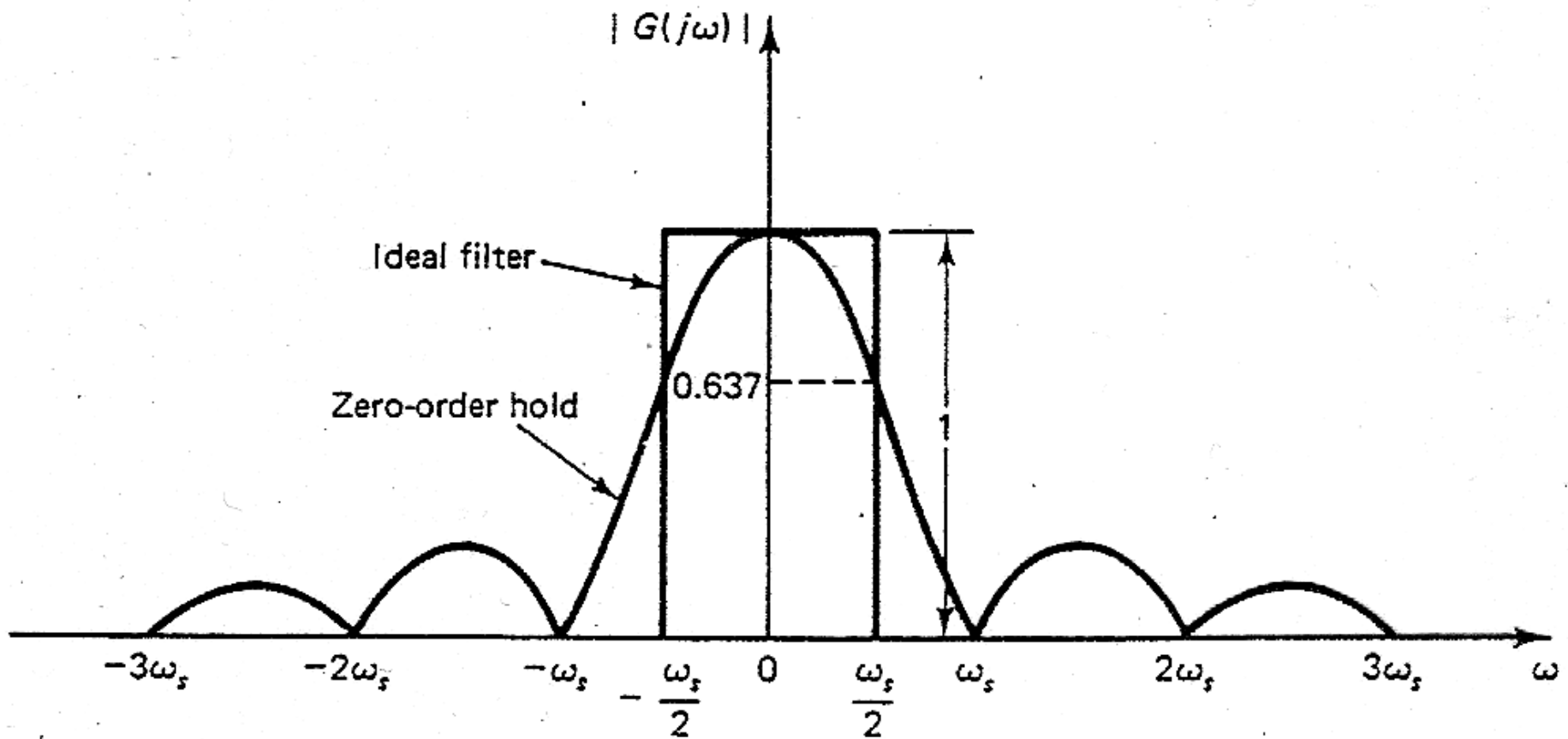
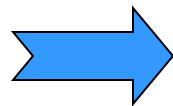


Figure 3-16 Comparison of the ideal filter and the zero-order hold.

- See Summary (P124)
- Problems: 3-1, 3-2, 3-3, 3-4(a), 3-5
3-9, 3-12, 3-15



Test Bank